

1) Measures of Central Tendency

i) Mean:

Mean is the average value of a set of data calculated by dividing the sum of all values by the total number of values.

$$\text{Mean} = \frac{\text{Sum of all values}}{\text{Number of values}}$$

$$\text{Mean} = \frac{\sum x}{n}$$

Ex: Describe the performance of students in python module?

38 40 42 44 46 50 54 56 58

x_1 x_2 x_3 x_9

$$\begin{aligned} \text{Mean} &= \frac{\text{sum of marks}}{\text{No. of students}} = \frac{\sum x_i}{n(9)} (x_1 + x_2 + x_3 + \dots + x_9) \\ &= \frac{428}{9} = 47.5 \end{aligned}$$

ii) Median:

Median is the middle value in a data set when the values are arranged in descending or ascending order.

If n is odd:

$$\text{Median} = \left[\frac{n}{2} + 1 \right]^{\text{th}} \text{ observation}$$

If n is even:

$$\text{Median} = \left[\frac{n}{2}, \frac{n}{2} + 1 \right]^{\text{th}}$$

Ex:

42	44	46	48	50	52	54	56	58
1	2	3	4	5	6	7	8	9

$$n=9$$

$$\Rightarrow \left[\frac{n}{2} + 1 \right] = \left[\frac{9}{2} + 1 \right] = [5.5] = 5$$

$$\text{Median} = 50$$

ii) Mode:

Mode is the value that appears most frequently in a dataset.

Ex:

40	45	50	50	50	55	60
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$$\text{Mode} = 50$$

a) What it means to find the center of data.

A single value that represents the entire dataset and shows where most values are concentrated.

b) Difference between Mean and Median.

$$\text{Mean} = \frac{\sum x}{P}$$

Median

1) Average of all values.

1) Middle value of ordered data.

2) Formula: $\frac{\sum x}{n}$

2) Middle observation.

3) Affected by outliers.

3) Not affected by outliers.

4) Best for normally distributed data.

4) Best for skewed data.

5) Used when data has no extreme values.

5) Used when data contains extreme values.

c) When businesses prefer Median over Mean, especially for skewed data.

Businesses prefer the median when the data is skewed or contains outliers, because extreme values can greatly affect the mean while the median remains stable.

2) Measures of Dispersion

i) Range: The difference between the highest and lowest values.

formula: $\text{Range} = \text{Max value} - \text{Min value}$

Ex:

Data: 10, 15, 20, 25, 30

$$\text{Range} = 30 - 10 = 20$$

ii) Variance: Measures the average squared deviation of each value from the mean.

$$\text{formula: } s^2 = \frac{\sum (x - M)^2}{n}$$

Ex:

Data: 2, 4, 6, 8, 10

$$\text{Mean} = \frac{2 + 4 + 6 + 8 + 10}{5} = \frac{30}{5} = 6$$

x	2	4	6	8	10
(x-6)	2-6	4-6	6-6	8-6	10-6
(x-6) ²	-4	-2	0	2	4
(x-6) ²	16	4	0	4	16

$$\sum (x - \bar{x})^2 = 16 + 4 + 0 + 4 + 16 = 40$$

$$\text{Variance} = \frac{40}{5} = 8$$

iii) Standard Deviation: Measures the average spread of data from the mean

formula:

$$\sigma = \sqrt{\frac{\sum (x - \bar{x})^2}{n}}$$

Ex:

Data: 2, 4, 6, 8, 10

$$\text{Mean} = \frac{2+4+6+8+10}{5} = 6$$

	2	4	6	8	10
$x - \bar{x}$	-4	-2	0	2	4

$(x - \bar{x})^2$	16	4	0	4	16
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$$\sum (x - \bar{x})^2 = 40$$

$$\text{Variance} = \frac{40}{5} = 8$$

$$\text{Standard deviation} = \sqrt{8} \approx 2.83$$

iv) Quartiles:

Q_1 : 25% of data lies below it.

Q_2 : 50% of data lies below it. (Median).

Q_3 : 75% of data lies below it.

Ex:

Data: 10, 20, 30, 40, 50, 60, 70, 80

$$Q_2 = \text{Median} = \frac{40+50}{2} = 45$$

Q_1 : Lower half: 10, 20, 30, 40

$$Q_1 = \frac{20+30}{2} = 25$$

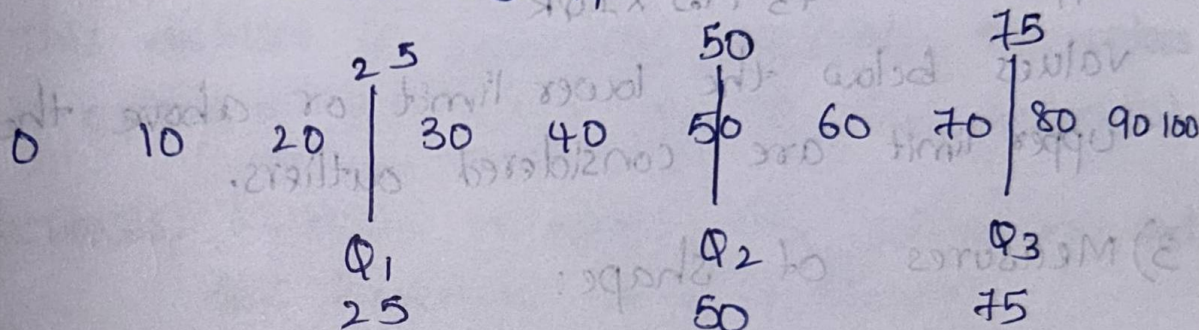
Q_3 : upper half: 50, 60, 70, 80

$$Q_3 = \frac{60+70}{2} = 65$$

v) IQR (Inter Quartile Range): IQR measures the spread of the middle 50% of the data.

formula: $IQR = Q_3 - Q_1$

Ex:



$$\begin{aligned} IQR &= Q_3 - Q_1 \\ &= 75 - 25 \\ &= 50 \end{aligned}$$

a) Why knowing only the average is not enough.

The average gives only the center of the data but does not show how spread out the values are. Two datasets can have the same average but very different variability.

b) How dispersion measures help understand

consistency, variability, and risk.

consistency: Lower spread means more consistent data.

Variability: Higher spread means greater variation.

Risk: Higher variability often indicates higher risk in business and finance.

c) How IQR is used to detect outliers.

The Inter Quartile Range is used to identify outliers in a data set.

formula:

$$IQR = Q_3 - Q_1$$

$$\text{Lower limit} = Q_1 - 1.5 \times IQR$$

$$\text{Upper limit} = Q_3 + 1.5 \times IQR$$

values below the lower limit or above the upper limit are considered outliers.

3) Measures of shape:

* skewness : skewness measures the symmetry of a data distribution.

i) Left-skewed distribution: The tail extends to the left, and most values are concentrated on the right.

$$\boxed{\text{Mean} < \text{Median} < \text{Mode}}$$

ii) Right-skewed distribution: The tail extends to the right, and most values are concentrated on the left.

$$\boxed{\text{Mean} > \text{Median} > \text{Mode}}$$

iii) Symmetric Distribution: A distribution where both sides are mirror images of each other.

$$\text{Mean} = \text{Median} = \text{Mode}$$

Ex:

Data: 10, 20, 30, 40, 50

$$\text{Mean} = \frac{10+20+30+40+50}{5} = 30$$

Median = 30

Mode = None

This dataset is symmetric because the values are evenly distributed around the center (30).

* Kurtosis: Kurtosis measures the peakedness and tail heaviness of a distribution.

i) Normal distribution: A normal distribution has a moderate peak and moderate tails.

Ex: Data: 60, 65, 70, 75, 80, 75, 70, 65, 60

ii) Heavy-tailed distribution:

A heavy-tailed distribution has a higher peak and heavier tails than a normal distribution, meaning it contains more extreme values.

Ex:

Data: 10, 11, 12, 13, 14, 15, 80.

The value 80 is an outlier, indicating a heavy-tailed distribution.

iii) Light-Tailed Distribution

A light-tailed Distribution has a flatter peak and lighter tails than a normal distribution, meaning extreme values are less likely.

Ex:

Data: 48, 49, 50, 50, 50, 51, 52.

The values are close to the center with no extreme values, indicating a light-tailed distribution.